Unit 4

AI-assisted Analysis of Physics Data using Supervised Learning in R

Physics Data Acquisition and Preparation; AI-assisted Data Analysis in R; Supervised Learning for Physics; Model Evaluation and Interpretation; Mini Project using Experimental or Publicly Available Physics Data.

Project Workflow

1. Collect or obtain a physics dataset (30–100 observations).
2. Use AI tools to inspect, clean, and preprocess the dataset.
3. Import the dataset into R.
4. Split the dataset into training and testing sets.
5. Train one or more supervised learning models.
6. Evaluate model performance using suitable metrics.
7. Interpret the results in the context of the underlying physics.

Example mini-projects

You can choose from projects such as:

Project	Model*	Parameter Prediction
Pendulum		Time period
Hooke's law		Extension
RC discharge		Voltage
Cooling experiment		Temperature
Photoelectric effect		Stopping potential
Ohm's law		Voltage
Mechanical oscillations		Frequency
Gas law measurements		Pressure or Volume

*The choice of model is left open.